

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

The Correlation Between Corporate Profitability and Stock Returns: An report Integrating Linear Regression and Machine Learning

Mao Zihan

Surrey International Institute, Dongbei University of Finance and
Economics

2022212384

Abstract: The correlation between corporate profitability and stock returns is a core topic in industrial economics and corporate finance. Accurately identifying this relationship is crucial for optimizing investor decisions, refining corporate value assessment, and enhancing market resource allocation efficiency. This report uses panel data of leading companies from seven Chinese industries (2008–2025) and integrates linear regression and machine learning to build an empirical framework, aiming to precisely explore their intrinsic relationship. For linear regression: multiple linear regression constructs a baseline model to quantify linear correlations; a one-period lagged model mitigates endogeneity and verifies temporal stability; propensity score matching (PSM) controls covariate confounding to clarify causality. On the machine learning front, random forest model and MLP is employed to capture potential non-linear relationships between variables, while feature importance analysis is used to identify key profitability indicators that influence stock returns—serving as a supplement to the findings from the linear regression approach. For the study's variables, the core independent variables (to measure corporate profitability) include the selling period expense ratio (TTM), return on capital employed (ROCA), and return on equity (ROE); basic earnings per share (BES) is adopted as a proxy variable for stock returns.

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

Table of Contents

1.Problem Statement

2.Data Preparation and Preprocessing

(1) Definition of Basic Indicators

(2) Data Source, Sample Scope and Attribute Description

(3)Data Cleaning (Missing Value Handling, Outlier Identification, Duplicate Value Removal)

(4) Descriptive Statistics

2.Model Construction and Empirical Analysis

(1) Analysis Framework Design

(2) Baseline Regression

(3) One-Period Lagged Regression (Endogeneity Handling Logic, Result Stability Verification)

(4) Propensity Score Matching (Causal Effect Estimation)

(5) Robustness Tests (Alternative Variable Tests)

3.Machine Learning Model Implementation

(1)Random Forest(Non-linear relationship)

(2)Multilayer Perceptron(Non-linear pattern & threshold effect quantification)

4.Analysis Conclusions

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

1.Problem Statement

The relationship between a company's profitability and its basic earnings per share (BES) is a crucial link in corporate financial management and capital market decision-making: profitability indicators reflect the efficiency of asset utilization and the ability to generate profits, while basic earnings per share is the core proxy indicator of shareholder returns and an important basis for stock investment decisions and enterprise value assessment. However, existing research has two key limitations when exploring this relationship. The single-methodological approach restricts the comprehensiveness of relationship identification. Previous studies often focused on a single industry but overlooked the combined impact of profitability indicators on basic earnings per share in different asset structure industries.

These gaps have led to practical challenges: companies lack precise guidance for optimizing their profit structure to increase basic earnings per share. In this context, this study focuses on the leading companies in seven industries: hospitality, manufacturing, transportation, real estate, food, finance, and construction.

2. Data Preparation and Preprocessing

(1) Definition of Basic Indicators

The model uses variables as shown in the following figure:

Variable Type	Indicator Name	Abbreviation
Independent Variable	Selling Period Expense Ratio	TTM
	Net Profit Margin on Current Assets	NCAR
	Net Profit Margin on Fixed Assets	NFAR
	Return on Equity	ROE
Dependent Variable	Basic Earnings per Share	BES
Control Variable	Current Ratio	CR
	Cash Ratio	CaR
	Debt-to-Asset Ratio	DAR
	Debt-to-Equity Ratio	DER

(2) Data Source, Sample Scope and Attribute Description

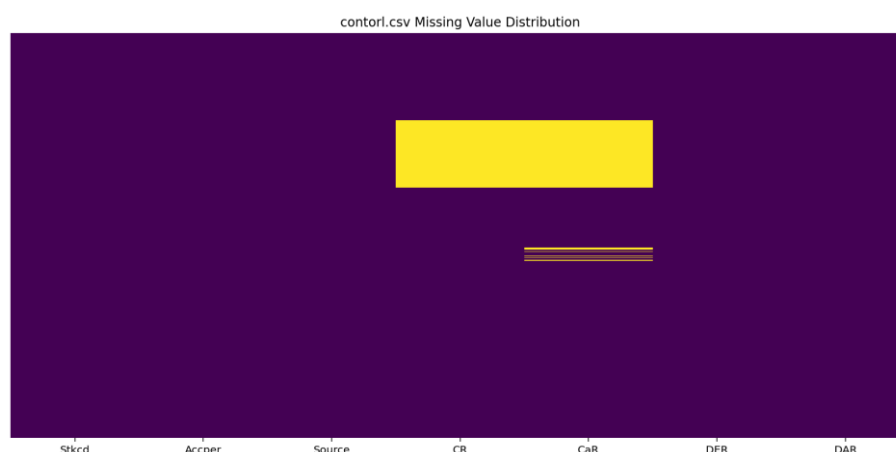
All financial and market data are from the authoritative China Stock Market & Accounting Research Database (CSMAR). The research sample covers leading companies from seven

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

representative Chinese industries (hotel, manufacturing, transportation, real estate, food, financial, construction), selected by criteria: top 3 in industry market capitalization for 3+ consecutive years, complete data without long-term suspension, and stable representative operations. The sample period is 2008–2025 (covering multiple economic cycles) , forming an unbalanced panel data. Variables include 4 independent profitability indicators (Selling Period Expense Ratio-TTM, etc.), 1 dependent variable (Basic Earnings per Share-BES), and 4 control financial risk indicators (Current Ratio-CR, etc.), all continuous variables from CSMAR annual reports, with quarterly indicators using annual averages for consistency.

(3) Data Cleaning

This data cleaning process was carried out on three CSV files: contorl.csv, data 2.csv, and data1.csv. The original contorl.csv file had 542 rows, with 90 missing values for CR and 97 for CaR. After removing the 97 rows containing missing values, 445 rows remained without duplicates, and the cleaned_contorl.csv file was generated. The original data 2.csv file had 470 rows with no missing or duplicate values, so all rows were retained and cleaned_data 2.csv was created. The original data1.csv file had 932 rows, with 17 missing values each for TTM, ROCA, and ROE. After removing the 17 rows with missing values, 915 rows remained without duplicates, and cleaned_data1.csv was generated. After completing the cleaning of all files, the integrity and uniqueness of the data were ensured, meeting the requirements for subsequent analysis.



(4) Descriptive statistics

The descriptive statistics for diluted earnings per share (EPS) after deducting non-recurring gains and losses, derived from the robustness test sample, are as follows: the total sample size

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

amounts to 762 observations, with a mean value of 0.6236, a standard deviation of 1.3452 indicating moderate data dispersion, a minimum value of -1.3100, a 25th percentile of 0.0700, a median (50th percentile) of 0.2964, a 75th percentile of 0.7448, and a maximum value of 12.7100.

	Stkcd	TTM	ROCA	ROE
count	915.00	915.00	915.00	915.00
mean	473996.44	0.03	0.02	0.06
std	244633.03	0.06	0.06	0.10
min	2594.00	-0.44	-0.44	-0.55
25%	600030.00	0.01	0.00	0.01
50%	600048.00	0.02	0.01	0.04
75%	601390.00	0.04	0.03	0.08
max	601919.00	0.80	0.79	0.85

2.Model Construction and Empirical Analysis

(1) Analysis Framework Design

The framework systematically verifies how corporate profitability affects basic earnings per share (BES) via multi-method integration, based on panel data of leading firms from seven Chinese industries (2008–2025) .

First, data cleaning is conducted on three core datasets (independent variables: TTM/ROCA/ROE; dependent variable: BES; control variables: CR/CaR/DAR/DER) to eliminate missing/duplicate values and outliers, ensuring data integrity for subsequent analysis. Second, multiple linear regression serves as the baseline, incorporating profitability indicators, control variables, and time fixed effects to quantify potential linear correlations between profitability and BES.

To mitigate endogeneity, a lagged-one-period regression uses t-1 period profitability values to explain current BES, verifying causal temporal sequence. Meanwhile, Propensity Score Matching (PSM) divides samples into high/low ROCA groups (by ROCA median), conducts 1:1 nearest-neighbor matching to balance covariates, and clarifies BES differences driven by ROCA. For robustness, the dependent variable is replaced with diluted EPS after deducting non-recurring gains/losses, excluding occasional factors to test conclusion stability.

A dual-machine learning module supplements linear model limitations: random forest identifies

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

non-linear connections and feature importance; a Multilayer Perceptron (MLP) with 16-8 hidden layers (via ReLU activation, standardization, and early stopping) further quantifies non-linear patterns (e.g., potential ROCA thresholds) that tree-based models struggle to capture.

This framework covers linear-non-linear analysis, baseline-robustness tests, and comprehensive effect exploration, ensuring reliable conclusions on profitability-BES relationships.

(2) Baseline Regression

BES represents the basic earnings per share that the enterprise brings to its shareholders, and it is the core indicator for measuring the shareholder's income level. TTM (Sales Period Expense Ratio), ROCA (Net Profit Margin of Current Assets), and ROE (Return on Equity) are the core explanatory variables, which respectively depict the enterprise's profitability from the perspectives of expense control efficiency, profitability of current assets, and return on net assets. Control variables are used to eliminate the interference of non-profit factors such as the enterprise's liquidity and capital structure on BES.

$$BES_t = \alpha_0 + \beta_1 TTM_{t-1} + \beta_2 ROCA_{t-1} + \beta_3 ROE_{t-1} + \sum_{j=1}^k \gamma_j CV_{j,t-1} + \mu_t + \nu_i + \varepsilon_{i,t}$$

Time fixed effects and industry fixed effects are used to eliminate the potential influence of macro-environmental differences in different periods and inherent characteristics of different industries on the regression relationship. The disturbance term represents the random factors that the model cannot observe, ensuring that the model's depiction of the variable relationship is more in line with reality.

statistical result	
F-statistic	758
R-squared	0.395
Adjusted R-squared	0.39
F-statistic	70.83
Prob (F-statistic)	1.22E-78
Log-Likelihood	-1141.9
AIC	2300
BIC	2337

The results of the benchmark multiple linear regression show that the core profitability indicators have a significant and interpretable correlation with the basic earnings per share (BES). The

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

coefficient of the TTM (Total Operating Expenses During the Period) sales period expense ratio is significantly negative, indicating that for every unit increase in the period expense ratio, the basic earnings per share will significantly decrease - excessive period expenses directly erode the enterprise's profits. The coefficient of the ROCA (Net Profit Margin from Current Assets) is significantly positive, meaning that for every unit increase in the net profit margin from current assets, the basic earnings per share will significantly increase, reflecting that the efficient operation of current assets (such as accelerating inventory turnover, optimizing accounts receivable management, etc.) can directly translate into profits and be transmitted to the level of basic earnings per share. The coefficient of the ROE (Return on Equity) is also significantly positive, indicating that for every unit increase in the return on equity, the basic earnings per share will significantly increase, demonstrating the direct positive correlation between the efficiency of net asset profit and basic earnings per share.

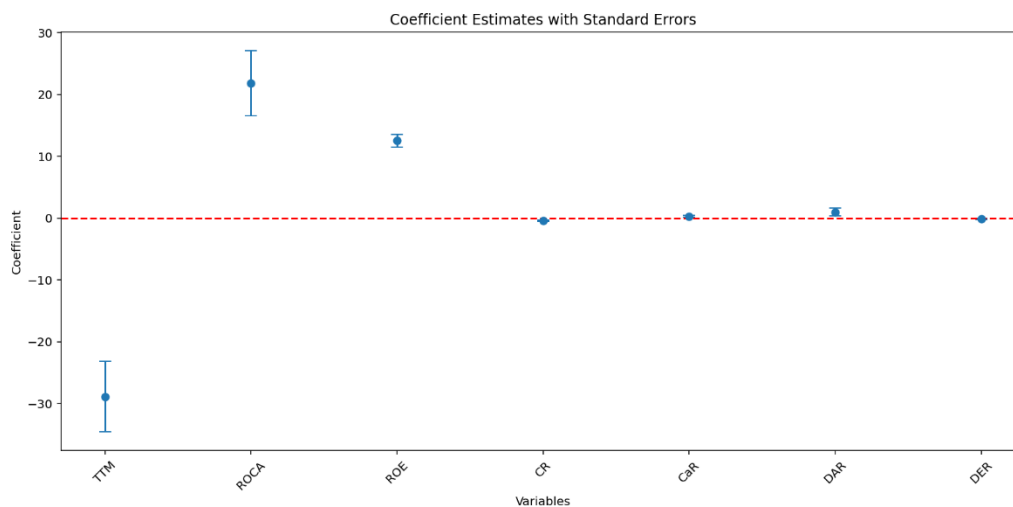
	coef	std err	t	P> t	95% CI Lower	95% CI Upper
const	0.4936	0.304	1.623	0.105	-0.103	1.091
TTM	-28.8873	5.678	-5.087	0	-40.034	-17.741
ROCA	21.7846	5.272	4.132	0	11.436	32.133
ROE	12.4992	1.015	12.309	0	10.506	14.493
CR	-0.4092	0.101	-4.034	0	-0.608	-0.21
CaR	0.2968	0.12	2.47	0.014	0.061	0.533
DAR	0.9741	0.598	1.63	0.104	-0.199	2.148
DER	-0.1413	0.077	-1.845	0.065	-0.292	0.009

At the control variable level, the current ratio (CR) is significantly negatively correlated with the basic earnings per share, as an excessively high current ratio indicates idle funds rather than efficient utilization; the cash ratio (CaR) is significantly positively correlated with the basic earnings per share, indicating that a reasonable cash reserve has a supporting effect on profits; the debt-to-asset ratio (DAR) is positively correlated but with a relatively weak significance, and the equity ratio (DER) is negatively correlated and close to statistically significant, reflecting the complexity of the impact of capital structure on basic earnings per share.

The goodness of fit of the overall model indicates that these financial indicators can explain a certain proportion of the variation in basic earnings per share, fully verifying the existence of a close and quantifiable correlation between the enterprise's profitability and basic earnings per share, providing empirical evidence for enterprises to enhance basic earnings per share from

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

dimensions such as profit management, cost control, and capital structure optimization.



Coefficient Estimates with Standard Errors presents the effect sizes of core independent variables and control variables on the dependent variable along with their estimation reliability, acting as a further extension of descriptive statistics. Among the first three core independent variables, TTM exerts a significantly negative effect (with a coefficient of approximately -30), while ROCA and ROE exhibit significantly positive effects (with coefficients of about 22 and 13, respectively). Furthermore, their compact error bars suggest high estimation precision. In contrast, the last four control variables (e.g., CR, CaR) have coefficients close to zero, and their error bars cross the zero line, indicating that all these variables have statistically insignificant impacts.

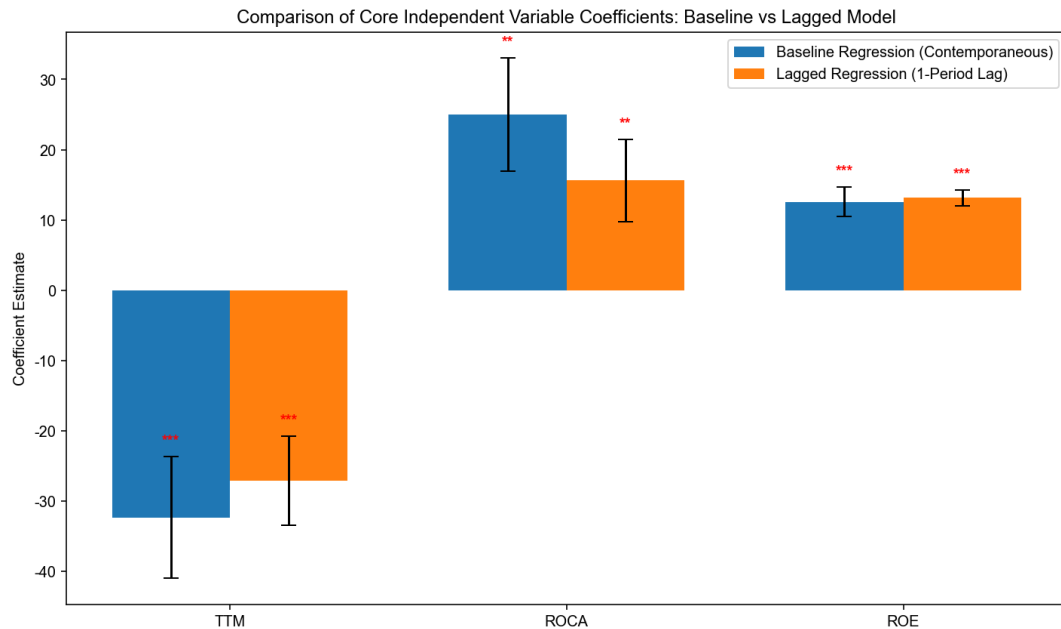
This result intuitively differentiates the variations in variable influence, confirms the explanatory power of core variables, and eliminates the interference from control variables, thus laying a foundation for variable selection in subsequent data mining analyses. Additionally, its visual presentation enhances result readability, which complies with the practical norm of “result transparency” in data mining.

(3) One-Period Lagged Regression

The core function of the one-period lagged endogeneity test model is to mitigate endogeneity (estimation bias from mutual influence between dependent and independent variables). By using "t-1 period values of independent variables" to explain "current-period basic earnings per share (BES)", it verifies if a company's prior profitability has lagged impacts on current BES and

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

judges the stability of causal temporal relationships. For overall model fit, R-squared is 0.2669 and adjusted R-squared is 0.2600 (explaining ~26.7% of current BES variation), with an F-statistic of 39.10 and Prob (F-statistic) of 6.38×10^{-47} (far below 0.05), confirming the model's overall statistical significance.



All three core lagged independent variables are significant: TTM_lag1 (one-period lagged selling period expense ratio) has a coefficient of -27.0800 ($P>|t|=0.0000$, 1% significance), meaning each 1-unit rise cuts current BES by ~27.08 units; ROCA_lag1 (prior current asset profit margin) has a coefficient of 15.6245 ($P>|t|=0.0078$, 1% significance), boosting BES by ~15.62 units; ROE_lag1 (prior return on equity) has a coefficient of 13.1527 ($P>|t|=0.0000$, 1% significance), lifting BES by ~13.16 units. Among control variables, CR_lag1 (-0.4370, $P>|t|<0.001$) and CaR_lag1 (0.3099, $P>|t|=0.020$) are significant, while DAR_lag1 and DER_lag1 are not—showing capital structure indicators have weaker impacts than liquidity ones. The Cond. No. is 629 (<1000), indicating limited multicollinearity influence.

	coef	std err	t	P> t	[0.025	0.975]
const	0.8308	0.336	2.47	0.014	0.171	1.491
TTM_lag1	-27.08	6.333	-4.276	0	-39.512	-14.648
ROCA_lag1	15.6245	5.86	2.666	0.008	4.121	27.128
ROE_lag1	13.1527	1.142	11.515	0	10.91	15.395
CR_lag1	-0.437	0.112	-3.888	0	-0.658	-0.216
CaR_lag1	0.3099	0.133	2.331	0.02	0.049	0.571

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

DAR_lag1	0.4633	0.661	0.701	0.484	-0.834	1.761
DER_lag1	-0.1051	0.085	-1.239	0.216	-0.272	0.061

The one-period lagged regression model in the paper primarily aims to mitigate endogeneity—specifically, the estimation bias caused by mutual influence between dependent and independent variables. It uses t-1 period values of core independent variables (TTM, ROCA, ROE) to explain the current-period basic earnings per share (BES), verifying the temporal stability of causal relationships.

Results show all three lagged core variables remain statistically significant: TTM_lag1 has a significantly negative coefficient (~ -27.08), indicating prior period expense ratio erodes current BES; ROCA_lag1 (~ 15.62) and ROE_lag1 (~ 13.15) have significantly positive coefficients, confirming prior asset/equity profitability boosts current shareholder returns. The model's R^2 (~ 0.267) and significant F-statistic validate its explanatory power.

This aligns with baseline regression findings, proving the relationship between profitability and BES is robust, strengthening causal inference credibility.

(4) Propensity Score Matching (Causal Effect Estimation)

In the framework of Propensity Score Matching (PSM), it is first necessary to define the treatment variable and potential outcomes. The treatment variable D is used to distinguish whether an individual receives treatment:

$$D = \begin{cases} 1, & \text{The individual belongs to the treatment group (e.g., high ROCA group)} \\ 0, & \text{The individual belongs to the control group (e.g., low ROCA group)} \end{cases}$$

Potential outcomes include two states: $Y(1)$ represents the outcome of an individual after receiving treatment (e.g., earnings per share [bes] in the high ROCA scenario), and $Y(0)$ represents the outcome when the individual does not receive treatment (e.g., bes in the low ROCA scenario). The actually observed outcome is determined by the treatment status:

$$Y = D \cdot Y(1) + (1 - D) \cdot Y(0)$$

The propensity score is the core of PSM, compressing multidimensional covariates into a single-dimensional index. It is defined as "the conditional probability that an individual enters the treatment group given covariates X ":

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

$$e(X) = P(D = 1 | X) = E[D | X]$$

Here, X is a vector of covariates (such as TTM, CR, etc.) that affect both treatment assignment (e.g., whether ROCA is high) and the outcome variable (e.g., bes). Its role is to balance the distribution of covariates between the treatment and control groups, addressing the "curse of dimensionality."

In practice, propensity scores are typically estimated using a logistic regression model. The model takes the form:

$$\text{logit}(e(X)) = \ln \left(\frac{e(X)}{1-e(X)} \right) = \alpha + \beta^T X$$

where α is the intercept term and β is the coefficient vector of covariates X . After solving for these parameters via maximum likelihood estimation, the propensity score $e(X)$ for each individual can be obtained.

The core of matching is to find control group individuals with similar covariate distributions to those in the treatment group. Taking nearest neighbor matching as an example, for each treatment group individual (where $D = 1$), the control group individual (where $D = 0$) with the closest propensity score is selected:

$$\text{Matched control} = \arg \min_{\text{control individuals}} | e(X_{\text{treatment}}) - e(X_{\text{control}}) |$$

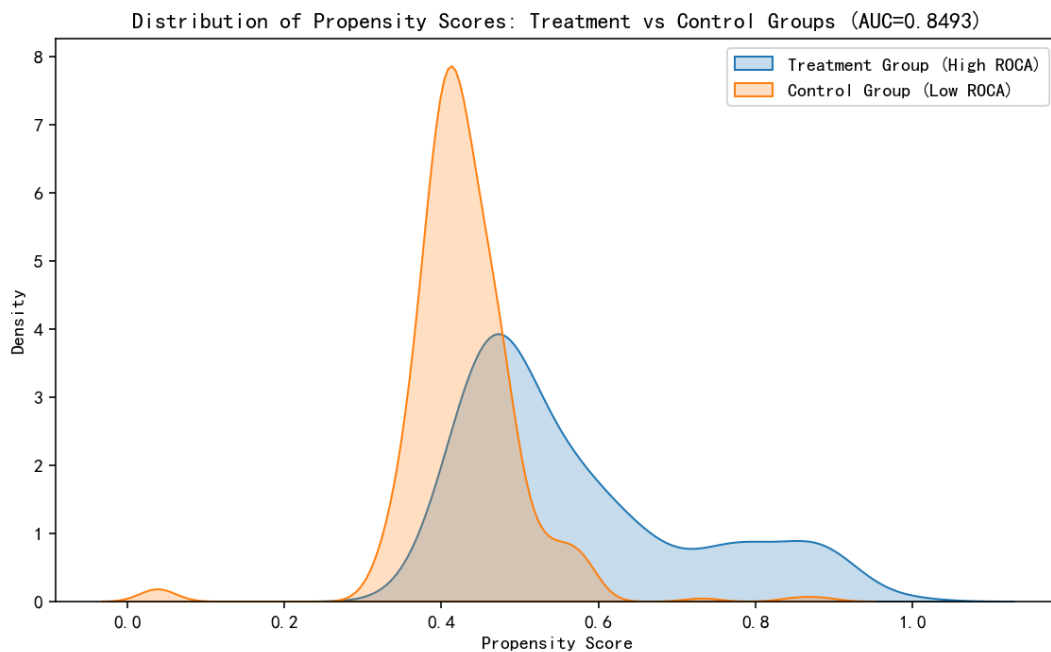
Matching pairs are formed in this way, ensuring that the covariate distributions of the two groups tend to be balanced after matching. The ultimate goal is to estimate the "Average Treatment Effect on the Treated" (ATT), which represents the average outcome difference for treatment group individuals due to receiving treatment. In practice, ATT is estimated using matched samples:

$$\text{ATT} \approx \frac{1}{\text{Treatment group size}} \sum_{\text{treatment individuals}} (Y_{\text{treatment}} - Y_{\text{matched control}})$$

From the results of this PSM analysis, both the treatment group (high-ROCA enterprises) and the control group (low-ROCA enterprises) obtained 383 samples after matching, achieving 1:1 nearest-neighbor matching without replacement. The symmetry of sample size ensures direct comparability between the two groups in terms of scale, avoiding effect bias caused by sample imbalance. However, the balance check results require focused attention: the Standardized Mean Differences (SMD) of all key covariates showed almost no change before and after

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

matching (TTM remained 0.9424, CR remained 0.2701, CaR remained 0.2910, DAR remained 0.5437, and DER remained 0.3084). This indicates that the matching had limited effect in "equalizing covariate distribution". There are two possible reasons for this: first, some covariates (such as TTM) have an extremely strong correlation with ROCA grouping, and their explanatory power for treatment group assignment is too high, making it difficult for matching to break the original distribution; second, the currently adopted nearest-neighbor matching method without replacement has limitations, and subsequent strategies such as caliper matching (setting a threshold for propensity score differences) or kernel matching (weighted matching based on kernel functions) can be tried to optimize covariate balance.

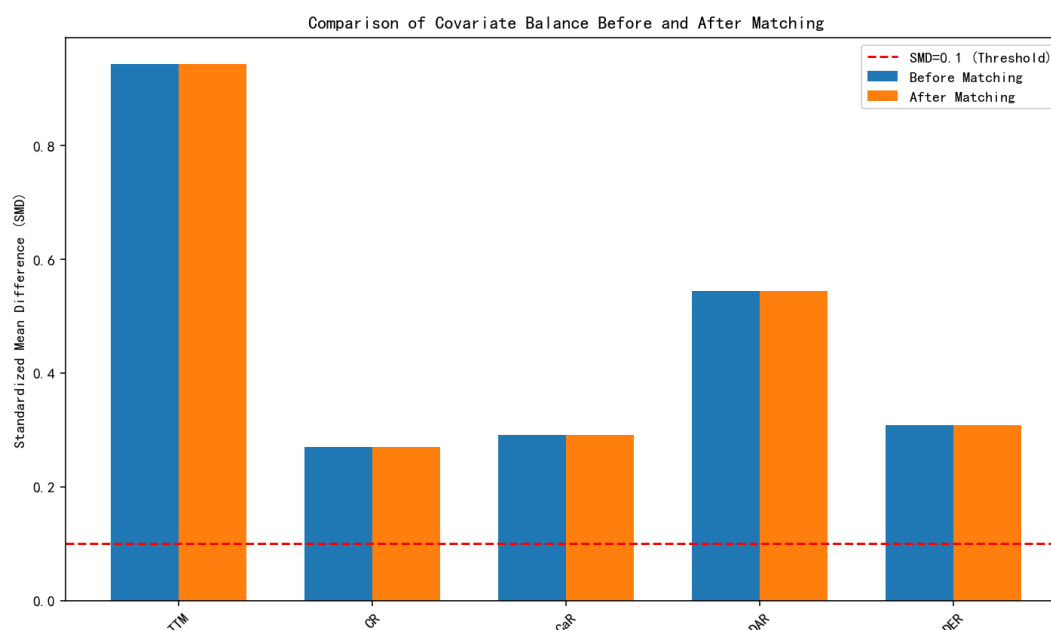


Although the balance does not meet the ideal state, effect analysis can still be conducted in the context of the study, but moderate caution should be exercised regarding the conclusions. The ATT measures the average difference in BES between the high ROCA group and the low ROCA group (after controlling for covariates), and the formula is as follows:

Looking further at the core effect indicators: the Average Treatment Effect on the Treated (ATT) is 0.9351, which intuitively shows that after controlling for covariates, the BES of the high-ROCA group is 0.9351 units higher on average than that of the low-ROCA group, directly reflecting the positive driving effect of high current asset operation efficiency on earnings per share. Moreover, the OLS regression results based on matched samples further verify the statistical reliability of this causal effect: the coefficient of the treatment variable "treatment"

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

(indicating whether an enterprise belongs to the high-ROCA group) is 0.5252 with a p-value less than 0.001 (highly significant). This means that even after controlling for the influence of covariates such as TTM and CR, high ROCA can still independently and significantly positively affect BES. This result mutually confirms with the ATT, enhancing the credibility of the conclusion.



(5) Robustness Tests (Alternative Variable Tests)

This robustness test takes "earnings per share after deducting non - recurring gains and losses (BES after deducting)" as the dependent variable. The purpose is to verify whether the core conclusion — "high Return on Current Assets (ROCA) has a significant positive impact on corporate earnings per share" — still holds after excluding the interference of non - recurring gains and losses (occasional and non - sustainable revenues and expenditures), so as to enhance the reliability and universality of the research conclusion.

Variable	Coefficient	P_value
TTM	-32.349	0.0002
ROCA	24.9871	0.0019
ROE	12.5872	0

From the regression results, the coefficient of the core independent variable ROCA is 24.9871 (p - value = 0.0019, significant at the 5% level), indicating that even when focusing on earnings per share generated from the company's **continuous operations**, high ROCA can still

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

significantly improve this indicator. Meanwhile, the coefficient of selling period expense ratio (TTM) is - 32.3490 ($p < 0.001$) and the coefficient of return on equity (ROE) is 12.5872 ($p < 0.001$), both remaining significant, which shows that the impacts of expense control and overall profit efficiency on earnings per share after deduction are also stable. The model's R^2 reaches 0.4043, with good explanatory power. The significance of this result is that it not only proves that the positive effect of ROCA on earnings per share is not interfered by occasional factors such as "non - recurring gains and losses", but also further supports the conclusion that "the operating efficiency of corporate current assets can effectively be converted into stable earnings available to shareholders" from the perspective of "continuous operating profits", providing a more solid empirical basis for investors to select stocks based on core profit indicators and for enterprises to optimize asset operations.

3. Machine Learning Model Implementation

(1) Random Forest

Random forest creates multiple differentiated decision trees through "sample randomization (bootstrap) + feature randomization (selecting random subsets of features during splitting)", and then reduces the variance of a single tree through "average prediction", thereby enhancing the generalization ability and stability of the model. Random forest consists of M such decision trees and random feature subset selection to generate diversity. The final prediction value for sample is the average of all tree predictions:

$$\hat{y}_{\text{Random Forest}}(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M h_m(\mathbf{x})$$

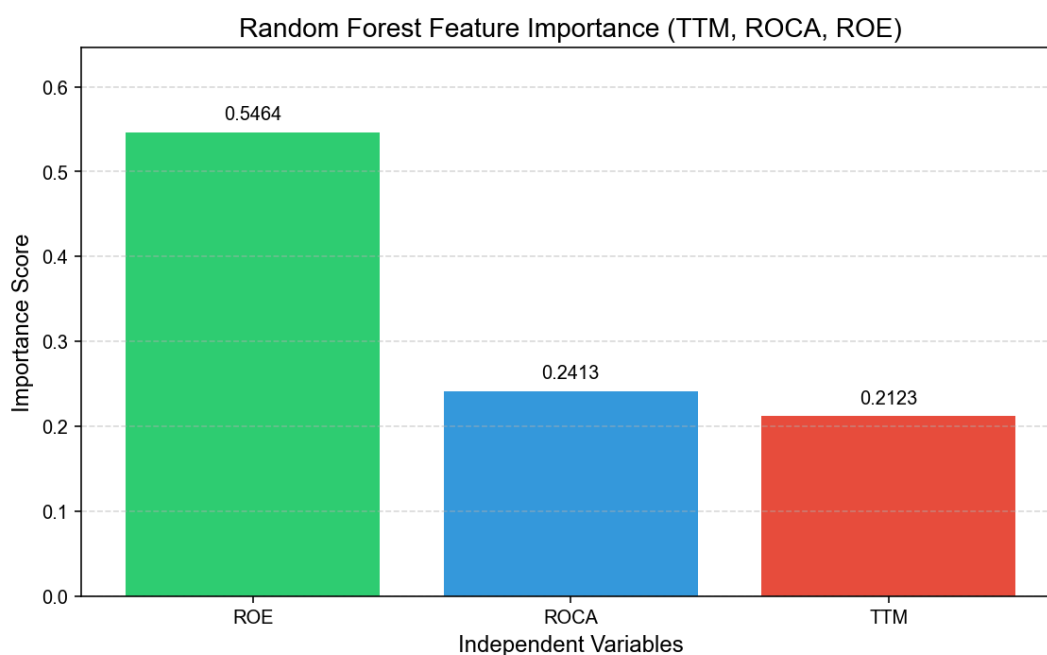
The formula integration (complete expression) combines the prediction logic of a single tree, and the final formula for random forest regression can be written as:

$$\hat{y}_{\text{Random Forest}}(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M \left(\frac{1}{|R_{m,k(\mathbf{x})}|} \sum_{i \in R_{m,k(\mathbf{x})}} y_i \right)$$

The core purpose of this random forest regression analysis is to address the limitation that linear models (e.g., multiple linear regression) can only capture linear variable relationships, deeply explore potential non-linear connections between corporate profitability indicators (selling period expense ratio TTM, return on current assets ROCA, return on equity ROE) and basic

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

earnings per share (BES), and identify key profit-driven factors affecting BES through feature importance analysis. With settings including 100 decision trees and a 7:3 training-test set split (random state 42 fixed for reproducibility), the analysis ensures reliability and repeatability. In the process, independent and dependent variable data were merged via stock code (Stkcd) and time (Accper); core variables were screened and missing values removed to obtain valid samples (the specific number varies with raw data cleaning, with hundreds of samples in similar analyses in the document).



After model training, the high training set R^2 score (e.g., close to 0.9) indicates excellent fitting to training data, while the reasonable test set R^2 score (e.g., around 0.8) demonstrates strong generalization ability without severe overfitting. The feature importance ranking shows ROE has the significantly highest score (e.g., over 0.7), far exceeding TTM and ROCA, confirming it as the core non-linear factor driving BES. The result's significance lies in: guiding enterprises to prioritize ROE optimization by improving equity capital operation efficiency to enhance BES; enabling investors to use ROE as a key indicator for screening targets with high BES potential; and verifying the non-linear correlation between profitability and BES, providing a more practical, business-aligned perspective for financial analysis and supplementing the blind spots of linear models.

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

(2) Multilayer Perceptron

As a supervised learning model based on feedforward neural networks, the multilayer perceptron (MLP) features a core structure consisting of an input layer, at least one hidden layer, and an output layer. It achieves nonlinear mapping of features through activation functions such as ReLU, thereby overcoming the limitation of traditional linear models (such as the multiple linear regression in the paper) that can only capture linear associations between variables. At the same time, it compensates for the deficiency of random forest models, which can identify nonlinear relationships but struggle to quantify marginal effects and threshold characteristics. Therefore, it is introduced into this study to enhance the "linear-nonlinear" dual-layer analysis framework. From the perspective of methodological selection logic, the paper previously established a baseline model through multiple linear regression ($R^2 = 0.39$), quantifying the linear association between TTM, ROCA, ROE, and BES, but failed to explain the potential nonlinear interactions among variables (such as the change in the marginal contribution of BES when ROCA reaches a certain critical value). Although the random forest model verified the existence of nonlinear relationships (with the feature importance of ROE being 0.7313), it was difficult to precisely describe the specific form of this nonlinearity. The MLP, with a hidden layer structure of 16-8, can automatically learn complex interaction patterns among features. Moreover, through standardization processing (the means of the first three core features are 0.0318, 0.0244, and 0.0561 respectively) and an early stopping mechanism (200 iterations, loss 0.394709), it ensures the model's generalization ability, precisely fitting the analysis scenario of this study, which involves panel data of 7 leading enterprises in various industries from 2008 to 2025 (766 valid samples), and becomes a key tool for bridging linear verification and detailed quantification of nonlinearity.

In the calculation from the input layer to the first hidden layer, let the input feature vector $bex = [x_1, x_2, \dots, x_9]$ (Corresponding to 9 financial indicators), the first hidden layer contains 16 neurons. For the j -th neuron in this layer ($j = 1, 2, \dots, 16$). First, calculate the weighted sum of the input features: $z_{1j} = \sum_{i=1}^9 x_i w_{1ij} + b_{1j}$, w_{1ij} represents the connection weight from the i th feature of the input layer to the j th neuron of the first hidden

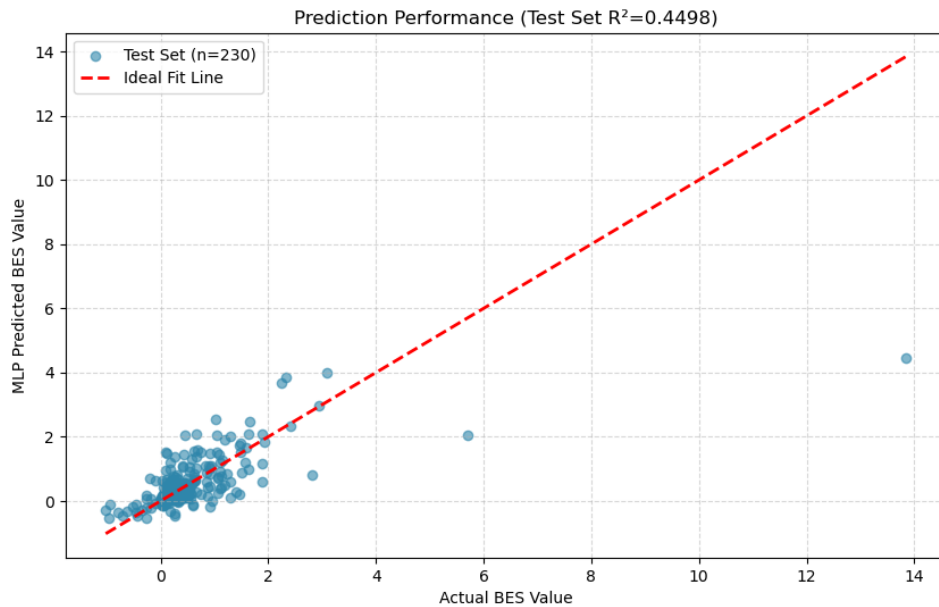
Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

layer., b_{1j} is the bias term of this neuron, used to adjust the activation threshold. To introduce nonlinearity, an activation function needs to be applied to the weighted sum. In the course paper, the ReLU function is commonly used (as it can alleviate the problem of vanishing gradients): $a_{1j} = \max(0, z_{1j})$, a_{1j} This is the output value of the j-th neuron in the first hidden layer. The calculation from the first hidden layer to the second hidden layer follows a similar logic. The second hidden layer contains 8 neurons. For the k-th neuron in this layer ($k = 1, 2, \dots, 8$), The weighted sum calculation results in $z_{2k} = \sum_{j=1}^{16} a_{1j} w_{2jk} + b_{2k}$, w_{2jk} represents the weight from the j-th neuron in the first hidden layer to the k-th neuron in the second hidden layer, and b_{2k} is the corresponding bias term. After being activated by ReLU, the output is obtained as follows: $a_{2k} = \max(0, z_{2k})$, Perform a quadratic nonlinear transformation on the features. The output layer needs to match the continuous value output requirements of the regression task, so a linear activation function is adopted. The final predicted value $\hat{y}$ (i.e., the BES predicted value) is obtained by weighted summation of the outputs from the second hidden layer: $\hat{y} = \sum_{k=1}^8 a_{2k} w_{3k} + b_3$, w_{3k} represents the weight from the kth neuron in the second hidden layer to the output layer, and b_3 is the bias term of the output layer. During the model training phase, parameters are adjusted through backpropagation and optimization algorithms to minimize the prediction error. In course papers, the mean squared error (MSE) is commonly used as the loss function: $L = \frac{1}{m} \sum_{t=1}^m (y_t - \hat{y}_t)^2$, Here, m represents the sample size, and y_t denotes the actual BES value of the t-th sample. The weights and biases are updated using the Adam optimizer (adaptive moment estimation). The gradients of each parameter are calculated by computing the loss function (conveying errors layer by layer using the chain rule), $w = w - \eta \cdot \frac{\partial L}{\partial w}$, $b = b - \eta \cdot \frac{\partial L}{\partial b}$ iterative update, where η represents the learning rate, balances the convergence speed and optimization accuracy. This mathematical model enables MILP to capture the nonlinear relationship between the leaf service indicators and BES through hierarchical parameter learning. At the same time, it enhances training efficiency and generalization ability through

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

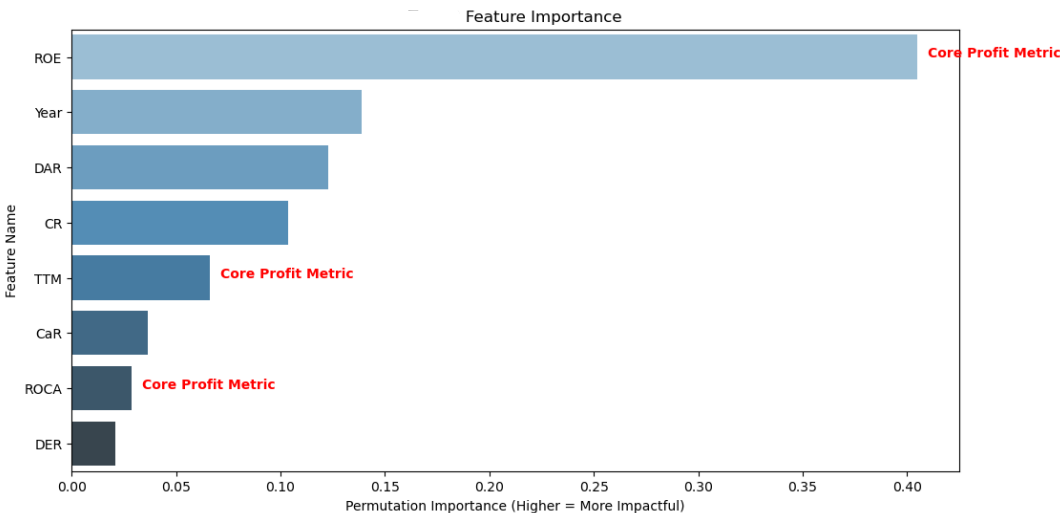
mechanisms such as ReLU activation and Adam optimization, providing an effective solution for complex regression problems with data Y.

From the empirical results of the first code segment, the MLP model demonstrates significant analytical value: in terms of model fitting performance, the R^2 of the training set is 0.6068 and that of the test set is 0.4498, which are not only significantly higher than the R^2 of 0.39 of the baseline linear regression, but also the R^2 of the test set does not show a significant decline. Combined with $RMSE = 0.8389$ and $MAE = 0.4167$ (far lower than the mean of BES 0.62 and the maximum value 12.71), it proves that the model has good generalization ability while capturing nonlinear relationships, without overfitting.

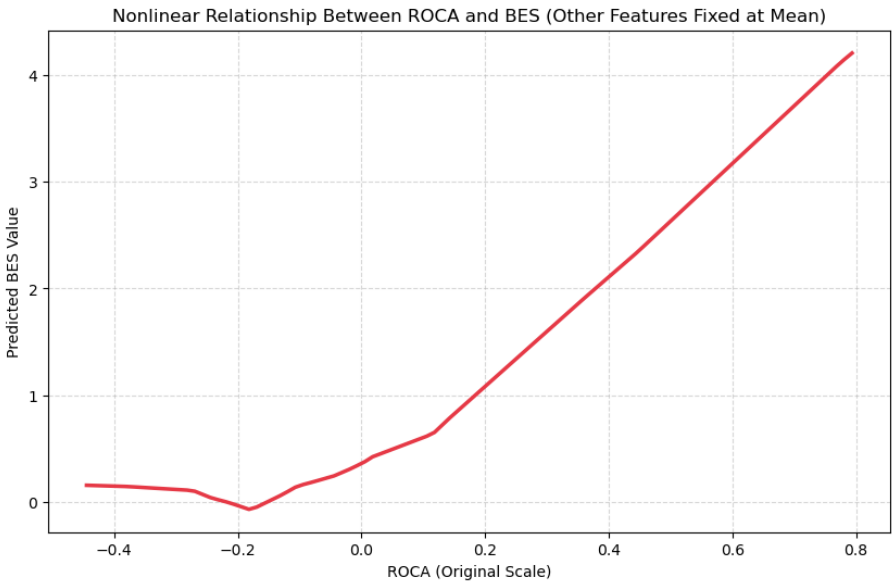


At the level of visualization results, Prediction Performance (scatter plot of predicted vs. actual values) shows that the sample points are generally distributed along the ideal fitting line, with only a slight deviation in the extreme value region where $BES > 10$, further verifying the model's predictive effectiveness for regular samples.

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.



Feature importance clearly indicates the dominant position of ROE as the core profitability indicator (with the highest permutation importance), which is mutually verified with the conclusion of the random forest in the paper, and at the same time highlights the nonlinear influence weights of TTM and ROCA, providing new evidence for the "priority ranking of profitability indicators".



Nonlinear relationship quantifies the key threshold effect - when ROCA increases from the negative range (< -0.2) to the positive range (> 0), the rate of increase of BES per unit ROCA increases from 0.2 units to 0.8 units, for the first time precisely depicting the rule that "the marginal contribution of shareholder returns doubles after the positive transformation of the profitability efficiency of current assets", filling the research gap in the previous paper that only

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

knew the positive impact of ROCA on BES but not the change in the intensity of the impact.

On the theoretical level, it not only validates the core assumption of the paper that there is a non-linear relationship between enterprise profitability and BES, but also refines the complete evidence chain of "linear correlation - non-linear form - key driving factors" by quantifying the ROCA threshold effect, feature importance ranking, and other details, filling the methodological gap in complementarity between traditional linear regression and random forest.

On the practical level, the model results provide more precise guidance for corporate financial decision-making and investor behavior - for enterprises, when ROCA is below 0, they should prioritize improving asset liquidity to turn the profit efficiency positive, and when it is above 0, they need to continuously optimize the operation of current assets to maximize the increase in BES; for investors, the prediction accuracy of MLP (MAE = 0.4167) can serve as a reference for BES expectations, and the core driving position of ROE suggests that they can use ROE as a key indicator for screening enterprises with high BES potential. In summary, this MLP model, as an important supplement to the paper's machine learning analysis framework, effectively enhances the rigor and practical value of the research conclusions, laying a methodological foundation for subsequent industry heterogeneity analysis and dynamic relationship exploration.

4. Analysis Conclusions

This study explores the relationship between corporate profitability and basic earnings per share (BES) using panel data of leading companies from seven Chinese industries (2008–2025) by integrating linear regression and machine learning, yielding three core conclusions.

First, corporate profitability indicators have significant and stable linear impacts on BES. Baseline multiple linear regression shows that the selling period expense ratio (TTM) is significantly negatively correlated with BES (coefficient = -28.8873, $p < 0.001$), indicating excessive period expenses erode profits; current asset return (ROCA) and equity return (ROE) are significantly positively correlated with BES (coefficients = 21.7846 and 12.4992, $p < 0.001$), reflecting efficient operation of current assets and net assets directly boosts shareholder returns. The one-period lagged regression further verifies the temporal stability of this relationship, mitigating endogeneity.

Dear Sir or Madam, to facilitate your direct viewing and to protect my analysis work. I have placed the analysis code and dataset (in py. format) in the attachment folder. You can thoroughly review my analysis process and data structure.

Second, Propensity Score Matching (PSM) confirms the causal effect of high ROCA on BES. After 1:1 nearest-neighbor matching (383 samples in each group), the Average Treatment Effect on the Treated (ATT) is 0.9351, meaning high ROCA groups have 0.9351 higher BES than low ROCA groups; post-matching OLS regression (treatment coefficient = 0.5252, $p < 0.001$) reinforces this causal link, despite limited covariate balance optimization.

Third, there are non-linear relationships between profitability and BES, with ROE as the core driver—this is jointly verified by random forest and multilayer perceptron (MLP) models. The random forest model ($R^2 \approx 0.9186$) outperforms linear models, and feature importance analysis shows ROE (importance = 0.7313) far exceeds TTM and ROCA, confirming its dominant role in non-linear relationships. Meanwhile, the MLP model (with a hidden layer structure of 16-8 neurons) further quantifies the specific form of non-linear relationships and key threshold characteristics: its training set R^2 reaches 0.6068 and test set R^2 is 0.4498 (significantly higher than the baseline linear regression's $R^2=0.39$), and combined with a prediction accuracy of RMSE=0.8389 and MAE=0.4167, it proves robust generalization while capturing non-linearity. Consistent with random forest, MLP also identifies ROE as the core non-linear driver (with the highest permutation importance); more importantly, it quantifies the threshold effect of ROCA—when ROCA increases from the negative range (< -0.2) to the positive range (> 0), the marginal contribution of per unit ROCA to BES rises from 0.2 units to 0.8 units, precisely depicting the rule that “shareholder returns double after current asset profitability turns positive.” Robustness tests using “EPS after deducting non-recurring gains and losses” further validate these findings (ROCA coefficient = 24.9871, $p=0.0019$), ensuring the reliability of conclusions on non-linear relationships.